CE / CS / SE 3354 · SOFTWARE ENGINEERING · FALL 2026

Project Deliverable P1

Proposal and Requirements

Weight 4% · Due Sunday 18 October 2026, 11:59 PM Central · Submit one PDF per team on Canvas

Field	Your entry
Project title	Study Room Reservation System for UTD
Team number and section	Team 07, Section 010
Team members (name · net ID)	Andrew Artiaga (axa220205), Artyom Gevorgyan (axg200155), Arjun Sharma (axs220320), Matthew Low (mjl230002), Ernie Hsiao (exh230014), Param Kathiravan (pxk230069)
Repository URL	https://github.com/matthewlow262-jpg/Study-Room-Reservation-System.git
Document version	1.0
Date	10/18/2026

0 Revision History and Contribution Record

Both tables below are graded. The contribution record carries the individual 20% of this milestone and must be consistent with your repository history.

Version	Date	Author	Summary of change
0.1	10/17/2026	All team members	Initial baseline for P1
0.2	10/17/2026	All team members	Drafting Sections 1-5
0.3	10/17/2026	Matthew Low	Finished Section 3.1, 4.1, 5
0.4	10/18/2026	Andrew Artiaga	Finished Section 1.2, GitHub draft upload, Section 2.2
1.0	10/18/2026	All team members	Finished all proposals and requirements

0.1 Individual contribution record

Team member	What this person produced for P1	Evidence in the repository
Matthew Low	Worked on part 1 introduction, section 2.6, section 3.1, section 4.1, section 4.2, section 5	Part 1 Intro, Section 3.1, Section 4.1, Section 4.2, Section 2.6, Section 5,
Andrew Artiaga	Worked on part 1, section 1.2, section 2.2, accepting/denying suggestions, GitHub management & contribution record	Initial upload, Part 1 · Record creation · Completed Section 2.2
Artyom Gevorgyan	Worked on section 2.1 and section 3.4	Section 2.1 Context Diagram, Section 3.4 Use Case Diagram
Arjun Sharma	Worked on part 2 product perspective(section 2.1), part 3 quality/constraint requirements and initial use cases(sections 3.2, 3.3 and 3.4) and part 4 risks(section 4.3)	Section 3.3, Section 4.3 Section 3.4 Section 3.2 Section 2.1
Ernie Hsiao	Worked on part 2.2 - 2.5, part 3.5	Sections 2.2-2.5 · Section 3.5
Param Kathiravan	Developed the two expanded use cases in Section 3.4 (UC-02: Reserve a Study Room and UC-05: Manage Room Listings)	Section 3.4 — Expanded Use Cases UC-02 and UC-05
Vimal Venkat	Never responded	

1 Introduction

1.1 Purpose of this document and of the system

This document is going to define the initial project proposal as well as the software requirements for the Study Room Reservation System for the University of Texas at Dallas. This document will describe the purpose, scope, stakeholders, constraints, requirements, and use cases. It is intended for the project team, instructor, and other stakeholders involved in reviewing, developing, and evaluating the project.

The Study Room Reservation System is going to give students a convenient way to find available study rooms on campus and reserve them for individual or group study. This system will allow students to search for rooms based on availability, time, and capacity. It will also let authorized personnel manage the room information.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Display a list of all study rooms with their locations and vacancies	Interacting with any physical locks and doors that require comet cards, keys, etc. Because this would require us to have access to external hardware and infrastructure that is only available to certain staff in UTD.
Allow students to search for rooms based off date, time, and capacity	Tracking real-time room occupancy through cameras or sensors because we are focusing on reserving rooms rather than monitoring the physical occupancy of the room
Display available time slots for a room	Integrating with UTD's current campus scheduling system or software because we do not have access to those systems
Allow students to make reservations for available time slots	Sending SMS notifications because we would require access to personal information from the students and that information is not currently required for the start of this project
Prevent overlapping reservations for the same room and time	Implementing UTD Duo authentication because that would require access to UTD's authentication system
Allow authorized administrators to manage the study room listings	Processing payments or charging students for room reservations. This feature is not necessary for the purpose of this system
Send email notifications regarding the status of the room reservation (confirmation/cancellation).	Integrate with external university services that have not been approved or made available to the team

Why the exclusions were excluded:

If we did include the exclusions, then the project would be much more difficult to create and to implement. The exclusions would require that we have access to external permissions and tools that go beyond the intended purpose of the program. By focusing on the aspects of the project that are in scope, we will be able to lower the complexity of the project and allow it to be used for its intended purpose.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
UTD	University of Texas at Dallas
Study room	A designated space for students to reserve for self-study or group study
Reservation	A booking that assigns a study room to whichever student booked it
System	An operating system designed and coded to allow students to make reservations
Availability	The status of a study room that shows if it is reserved or if it is free

1.4 References

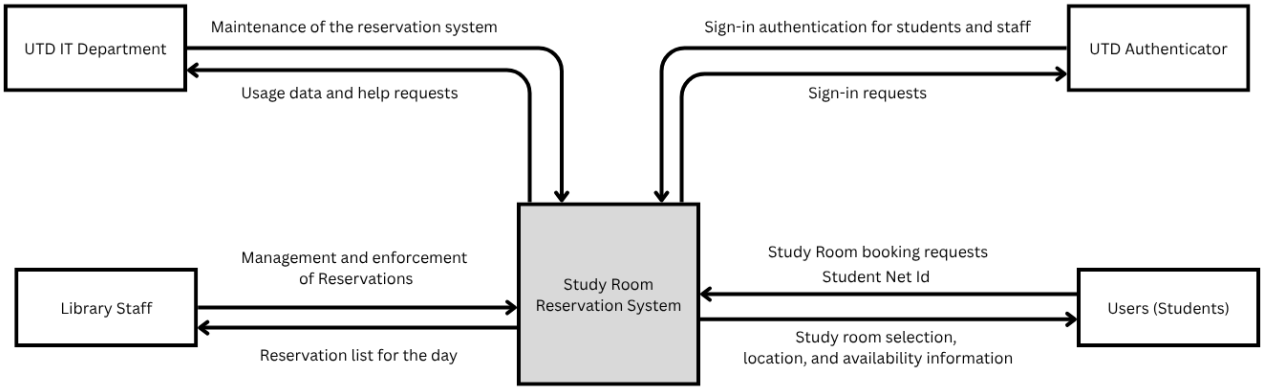
[R1] CE 3354 Fall 2026 course syllabus and P1 project template

[R2] UTD library policies; https://library.utdallas.edu/about-the-library/library-policies/

[R3] UTD policies on information security or privacy; https://policy.utdallas.edu/utdbp3096

2 Overall Description

2.1 Product perspective and context



Why the boundary sits here: Inside the box is everything we build: the room catalogue, availability search, reservation rules, admin tools for room listings, and the logic that triggers emails. Outside is everything we depend on but don't control: UTD's identity records, its email service, and the physical doors, locks and Comet Card readers. Only students and library staff use the system directly.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
Students	Users that use the system	- Reservation of rooms - Cancellation of reservations - Confirmation of reservation - Way to view availability	Based on our experiences as students, finding a study room in the library can be difficult at times, such as during midterms. These needs are currently based on our observations and assumptions.
Library Staff	Manage the study room area	- Room management capabilities, both physical and remote	We assume library staff are responsible for managing study room availability and resolving room-related issues. These assumptions have not

Stakeholder	Role in relation to the system	What they need from it	How you learned this
			yet been confirmed through interviews.
UTD IT Department	Create and maintain the system	- System security and user authentication integration - Database management capabilities - Compliance with UTD privacy policies	Based on our understanding of the technical requirements for a university reservation system. The specific authentication, infrastructure, and privacy requirements still need to be verified with UTD IT staff.
UTD Office of Budget and Finance	Fund the servers and system	- Proof that the system is being used - Proof that the system is helpful	We assume that funding decisionmakers would need evidence of system usefulness and operating costs before supporting continued funding.

2.3 User characteristics

Users: Students at UTD

They know: Where the study rooms are, study room hours, size of their party

They don't know: Which rooms are available, when rooms are available, how big the rooms are

User constraints: UTD ID

2.4 Constraints

The registration service must be hosted by UTD's servers. The data used to track students is personal and should remain on the UTD servers.

The system must comply with UTD’s rules and policies

The students must leave the study rooms at least 30 minutes before the library closes. This is to give the library staff time to clean up.

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	We depend on the library staff to manage the study rooms.	There will be no one to update or enforce the reservations.
A2	We assume the students and library staff have access to the internet.	They will not be able to place or change a reservation.
A3	An email delivery method will be available if email notifications are in scope	Confirmation and cancellation emails may not be deliverable

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Potentially feasible	The functions that the project needs involve room listings, time-slot searches, reservation records, and conflict checks. To tell if this is technically feasible, we will need to include a working proof of concept, repository code, and tests to check if the requirements can be met. Until then this will remain potentially feasible
Operational	Potentially feasible	Students can make reservations for available study rooms and the system will deal with conflicting reservations and if needed, the staff can manage the reservations as well. The team still needs to test the workflow with students and staff in the library
Schedule	Feasible	We have identified core functions that the system needs to be successful and have a milestone plan. Evidence of schedule feasibility will include assigned tasks to owners, completed milestones, recorded progress, etc. This way the schedule remains achievable
Economic / resource	Potentially feasible	The proposed scope does not include tools like paid reservation processing or integration with whole university systems. These conflicts could prove to be difficult because even if our project works it only remains potentially feasible because we cannot control things like integrating our system with the universities software.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall display the list of study rooms in its room listing	MUST	Students	A tester can confirm this by verifying that every room in the test dataset appears in the list
FR-02	The system shall display the location of each listed study room	MUST	Students	Verify that each of the listed rooms displays its location
FR-03	The system shall display the capacity of each listed study room	MUST	Students	Verify that each listed room displays its capacity
FR-04	The system shall allow students to search for rooms by reservation date.	MUST	Students	Search using a date and ensure that only rooms with availability are returned
FR-05	The system shall allow students to search for rooms by reservation time	MUST	Students	Search using a time range and make sure that the results match the availability of the rooms
FR-06	The system shall allow students to filter rooms by minimum capacity	SHOULD	Students	Enter a capacity and verify that each result either meets or exceeds it

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-07	The system shall display available time slots for a selected room	MUST	Students	Select a room and verify that the time slots match the rooms reservations
FR-08	The system shall allow a student to reserve an available time slot	MUST	Students	Submit a valid reservation and verify that the reservation is recorded
FR-09	The system shall prevent a reservation from overlapping with an existing reservation in the same room	MUST	Students	Attempt an overlapping booking and verify that it is rejected
FR-10	The system shall allow a student to cancel an existing reservation that they submitted	MUST	Students	Cancel a reservation and verify that its status is updated and the slot is now free
FR-11	The system shall display reservations made by the student requesting a reservation	COULD	Students	Verify that the student can see their own reservation records without seeing other students' reservation records
FR-12	The system shall allow authorized library staff to update the location and capacity of existing study room listings.	MUST	Library Staff	Log in as an authorized administrator, modify a room's location and capacity, save the changes, and verify that the updated information appears in the room listing.
FR-13	The system shall allow authorized library staff to view study room reservation usage data for a selected date range.	SHOULD	Library Staff	Select a date range as an authorized administrator and verify that the system displays reservation usage data for that range.

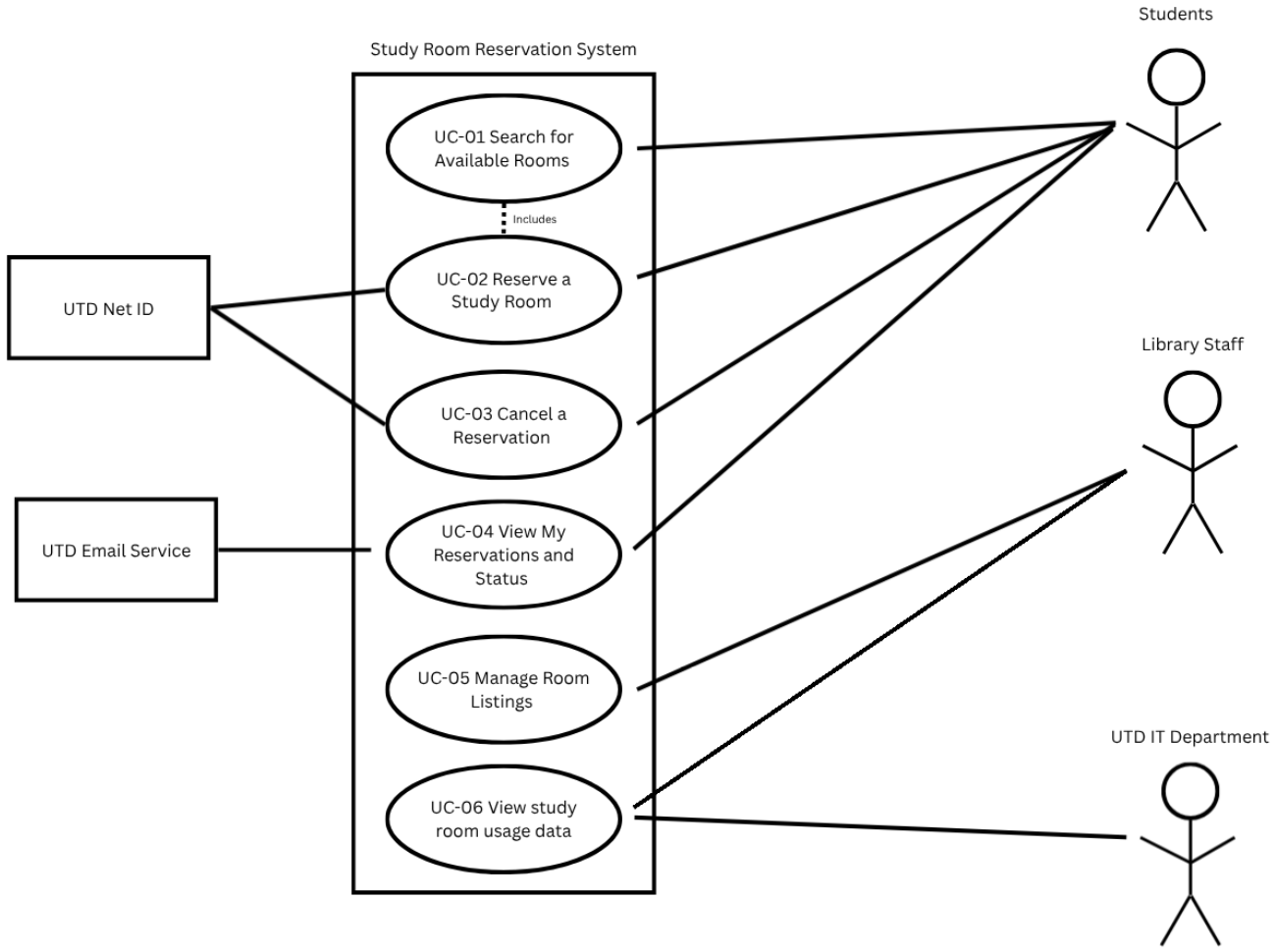
3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	95% of room searches (FR-04 and FR-05) will return results within 2 seconds for up to 25 concurrent users and 50 rooms.	MUST	Students	Load script of 200 searches run with 25 simultaneous users; the 95th-percentile response time is read from the script log.
QR-02	When 20 users attempt to reserve the same room and time slot at the same moment, exactly one reservation shall be created and 19 shall be rejected, in each of 100 repeated runs.	MUST	Students	Automated concurrency tests run 100 times; the database is checked after each run for a single confirmed reservation.
QR-03	At least 80 percent of first-time student testers (minimum 5 testers) shall find a room and complete a reservation in under 3 minutes without help, on a phone-size screen 360 pixels wide.	SHOULD	Students	Moderated usability test with a stopwatch and a task script; completion times and unaided successes are recorded.
QR-04	The system shall be available at least 99 percent of the time during library opening hours, measured over any 14-day period.	SHOULD	Students	A monitoring service requests the home page every minute during library opening hours; availability is computed from its log.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall be hosted on UTD-operated servers, and student personal data shall not be stored anywhere else.	UTD IT	Student identity data is personal, and UTD IT is responsible for keeping it on UTD servers.
CON-02	The system shall comply with UTD's rules and policies, including those on student privacy.	UTD IT	A university rule is not the teams to waive.

3.4 Initial use cases



Actors: Student and Library staff (administrator); supporting actor: UTD email service. System boundary as in 2.1.

UC-01 Search for Available Rooms

UC-02 Reserve a Study Room

UC-03 Cancel a Reservation

UC-04 View My Reservations and Status

UC-05 Manage Room Listings

UC-06 View study room usage data

Signing in is a precondition of UC-02 to UC-05, not a use case on its own: it is a step toward a goal, not a goal.

Expanded use case — copy this block for each of your two

Field	Content
Identifier and name	UC-02 Reserve a Study Room
Primary actor	Student
Supporting actors	None confirmed; email service
Preconditions — what must be true before it starts	The student is identified by the system, and an available study room and time slot exist
Trigger — what starts it	The student selects an available room and chooses to reserve it
Main success scenario — numbered steps	1. The student selects an available study room 2. The system displays available time slots 3. The student selects a time slot 4. The student submits the reservation request 5. The system checks for conflicting reservations. 6. The system records the reservation 7. The system displays confirmation to the student
Alternative and exception flows	1. If the selected time slot is unavailable, the system rejects the request and asks the student to select another slot. 2. If the student's identity cannot be confirmed, the system prevents the reservation. 3. If the reservation cannot be saved, the system reports the failure without confirming the booking
Postconditions — what is true when it succeeds	The reservation is recorded, and the selected room and time slot are no longer available for overlapping reservations.
Requirements realized	FR- 07, FR-08, FR-09

Expanded use case — copy this block for each of your two

Field	Content
Identifier and name	UC-05 Manage Room Listings
Primary actor	Library Staff (Administrator)
Supporting actors	None confirmed
Preconditions — what must be true before it starts	The library staff member is signed in with administrator permissions. The system has access to the current room listings.
Trigger — what starts it	The library staff member chooses to manage the study room listings.
Main success scenario — numbered steps	1. The administrator opens the room management section. 2. The system displays the current study room listings. 3. The administrator selects a room to update. 4. The system displays the room's current information.

	5. The administrator modifies the room's information, such as location or capacity. 6. The administrator submits the changes. 7. The system validates the updated information. 8. The system saves the changes and displays confirmation.
Alternative and exception flows	1. If the administrator enters invalid room for information, the system rejects the changes and requests corrections. 2. If the changes would conflict with existing reservations, the system rejects the update and informs the administrator. 3. If the administrator does not have the required permissions, the system denies access. 4. If the changes cannot be saved, the system displays an error and keeps the previous room information unchanged.
Postconditions — what is true when it succeeds	The updated room location or capacity is saved in the system and displayed in the room listings. Existing reservations remain unchanged.
Requirements realized	FR-01, FR-02, FR-03, FR-12

3.5 Requirements to use case traceability

Requirement ID	Use case(s) that realize it	If blank — why, and what you will do
FR-01	UC-01, UC-05	
FR-02	UC-01, UC-05	
FR-03	UC-01, UC-05	
FR-04	UC-01	
FR-05	UC-01	
FR-06	UC-01	
FR-07	UC-01, UC-02	
FR-08	UC-02	
FR-09	UC-02	
FR-10	UC-03	
FR-11	UC-04	
FR-12	UC-05	
FR-13	UC-06	

This reveals that our matrix has no current gaps, but it also shows which use cases are involved in most requirements. Because of this, we decided not to change anything, but will monitor any potential weak tests.

4 Project Plan

4.1 Team roles and working agreement

Team member	Role for P1 to P2	How the team makes a decision when it disagrees
Matthew Low	Requirements manager and repository maintainer	Discuss issues with the team, compare alternatives against the agreed requirements and record the decision
Andrew Artiaga	Scope, system context, and repository coordinator	Discuss disagreements with the team, compare options against the project scope and stakeholder needs and document decision
Arjun Sharma	Stakeholder and elicitation manager	Discuss disagreements with the team, review/evaluate stakeholder needs and document decision
Ernie Hsiao	Non-functional, constraints, and prototyping analyst	Discuss disagreements with the team, evaluate decisions against system performance and quality attributes, and document the decision
Artyom Gevorgyan	Verification and quality requirements analyst	Discuss disagreements with the team, compare and analyze options for reaching quality requirements, and document the decision
Param Kathiravan	Use case analyst	Discuss disagreements with the team, compare proposed solutions against functional requirements and use cases, and document the agreed decision.

4.2 Plan to P2

Date	What will be finished	Owner
Oct 18	Finalize and submit P1 requirements baseline	All team members
Oct 21	Review P1 feedback and identify problems and propose corrections to the overall project	All team members
Oct 25	Finalize stakeholder assumptions, reservation rules, and open questions	Arjun Sharma & Andrew Artiaga
Oct 28	Review requirements and use cases, update traceability and resolve gaps	Matthew Low
Nov 1	Complete P2 design-related deliverables	Ernie Hsiao
Nov 6	Complete the team review and resolve identified document issues	All team members
Nov 8	Submit the revised project documentation for P2	All team members

4.3 Risks

#	Risk	Likelihood	Impact	Mitigation — action and owner
R1	Student sign-in cannot be connected to UTD's identity service, so the	High	Medium	Arjun Sharma – Define a simple identification method that does not require UTD identity integration.

#	Risk	Likelihood	Impact	Mitigation — action and owner
	system cannot confirm who is booking.			
R2	Requirements change after this baseline and cause rework in P2 and P3.	Medium	High	Matthew Low – Record the requirement changes in the repository and assess the impact on scope and deadlines. Also make sure the team agrees to any changes proposed for the baseline
R3	A member is unavailable during exam season, and their work is delayed or missing.	Medium	Medium	Ernie Hsiao – Have each member document their progress and upload their work to GitHub regularly. Share status on tasks with the team and reassign unfinished tasks if necessary
R4	Two students are able to reserve the same room and time under load.	Low	High	Artyom Gevorgyan – Design a reservation conflict check that will recheck availability on reservations when submitted.
R5	Scope grows to include excluded items such as SMS, Duo, or door locks.	Medium	Medium	Andrew Artiaga – Keep a list of extra features whether they are used or not and decide with the team whether these features fit in the scope of the project or not.

4.4 Repository and tooling

https://github.com/matthewlow262-jpg/Study-Room-Reservation-System.git

Commit conventions and Branch: We have decided to focus on using the Main branch as this will contain all of the team's reviewed project files.

Access and permissions: Team members should have the repository permissions needed to access all the team's files.

How issues are used: We will mainly use GitHub to document our progress, and issues will be discussed by team members to come to a consensus.

4.5 Declaration of AI tool use

Tool	What it was used for	Who used it	How the output was checked
ChatGPT	Understanding what section 1.4 References needs and should be	Matthew Low	Compared the guidance with P1 project template and course requirements to come up with the three references
ChatGPT	Helped refine the project description and format the P1 contribution record, including GitHub commit links	Andrew Artiaga	Reviewed the wording for accuracy and that the GitHub commit links matched the project work

5 Open Questions

#	The question	Who can answer it	Needed by	What you assumed in the meantime
Q1	Will the system use actual UTD study room information or a simulated dataset?	Team members; UTD library staff	October 25, 2026	The system will use a simulated dataset if actual UTD room information is unavailable.
Q2	How will the students identify themselves when making, viewing, or canceling reservations?	Project team; UTD IT staff authentication is available	October 25, 2026	The initial version will use a simple project-level identification method and will not integrate with UTD Duo.
Q3	Will the system send email confirmations, and what email service can the team use?	Project team; the instructor or UTD IT staff	October 25, 2026	Email notifications are optional until the team confirms that an email service is an available option.

6 Baseline Declaration

We confirm that the requirements in section 3 represent our agreed baseline as of the submission date, and that every team member has read this document in full.

Team member	Signature or typed name	Date
Matthew Low	Matthew Low	10/17/2026
Artyom Gevorgyan	Artyom Gevorgyan	10/17/2026
Ernie Hsiao	Ernie Hsiao	10/17/2026
Arjun Sharma	Arjun Sharma	10/18/2026
Andrew Artiaga	Andrew Artiaga	10/18/2026
Param Kathiravan	Param Kathiravan	10/18/2026